

Technical Report: Customer Classification System using XGBoost

Executive Summary

I developed a binary classification system using **XGBoost** to segment wholesale customers into two categories: **Horeca (Hotel/Restaurant/Café)** and **Retail**. The model achieved **92% accuracy** using advanced machine learning techniques and cross-validation, identifying key factors driving customer purchasing profiles.

Methodology and Process

1. Data Preparation and Exploratory Analysis

- **Dataset:** Wholesale Customers Data (UCI Machine Learning Repository)
- **Volume:** 440 observations, 8 variables
- **Variables Analyzed:**
 - *Target:* Channel (Horeca vs. Retail)
 - *Features:* Region, Fresh, Milk, Grocery, Frozen, Detergents_Paper, Delicassen
- **Data Quality:** Complete dataset with no missing values
- **Descriptive Analysis:** Identified distributions and outliers in annual spending across categories

2. Feature Engineering and Preprocessing

- Transformed the target variable to [0,1] for XGBoost optimization
- Train-test split: 80%-20% with stratification
- Implicit normalization via XGBoost algorithm

3. XGBoost Model Development

This is only a technical summary; the full detailed version can be found in the Spanish document.

```
In [ ]: # Technical Implementation
model = xgb.XGBClassifier()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

4. Validation and Optimization

- **Cross-validation:** 10 folds
- **Evaluation Metrics:** Accuracy, Log Loss
- **Overfitting Analysis:** Train-test loss comparison
- **Feature Importance:** Weight-based analysis

Key Results

Model Performance

- **Accuracy:** 92%
- **Final Log Loss:** 0.249 (test set)
- **Prediction Distribution:**
 - 70.4% Horeca Customers
 - 29.5% Retail Customers

Feature Importance

1. **Grocery** (97 points) - Primary determining factor
2. **Detergents_Paper** (94 points) - Secondary key influence
3. **Delicassen** (92 points) - Third key predictor
4. **Frozen** (82 points)
5. **Fresh** (77 points)

Cross-Validation Results

The model demonstrated stability across iterations:

- Progressive reduction in training and test log loss
- Minimal difference between train-test loss (≈ 0.149)
- Stable convergence without significant overfitting

Technical Skills

Programming and Machine Learning

- **Python:** Pandas, NumPy, Scikit-learn, XGBoost
- **Machine Learning:** Binary classification, ensemble methods
- **Model Validation:** Cross-validation, metric analysis
- **Visualization:** Matplotlib, feature importance plots

Data Analysis

- **EDA:** Statistical analysis, data quality assessment
- **Feature Engineering:** Target transformation, data splitting
- **Model Interpretation:** Feature importance analysis
- **Business Insights:** Translating technical results into business value

Business Impact and Applications

Effective Segmentation

The model enables:

- Automatic customer channel identification with 92% accuracy
- Segment-specific marketing strategy optimization
- Personalized commercial offers based on purchasing patterns

Actionable Insights

- **Grocery, detergents, and delicatessen** are the top predictors
- Horeca customers represent 70% of the portfolio
- Potential to expand the model for sub-segmentation within each channel

Technologies: Python, XGBoost, Scikit-learn, Pandas, Matplotlib

Methodologies: Cross-validation, Feature Importance Analysis, Binary Classification

Outcome: 92% accuracy in automated customer segmentation